

REAL ANALYSIS

Real Number:

All numbers, natural, whole, integers, rational and irrational are real numbers. The set of real numbers is usually denoted by R .

$$R = Q \cup Q' = (-\infty, \infty)$$

Real Function:

A function $f(x)$ is said to be real function if range set $f(x)$ is real.

Countable Set :

A set is said to be countable if A is either finite or denumerable. A sets is said to be denumerable if there exists a bijection of N onto S . The sets, N , W , Z and Q are countable sets.

Supremum:

A set S is said to be bounded above if there exists $m \in R$ such that $x \leq m$ for all $x \in S$. Each such m is called upper bound of S . The least upper bound is called supremum of S .

Infimum:

A set S is said to be bounded below if there exists $t \in R$ such that $t \leq x$ for all $x \in S$. Each such t is called lower bound of S . The greatest lower bound is called infimum.

Following table shows the information of supremum and infimum of some subset of real numbers.

Set	Supremum	Infimum
ϕ	$-\infty$	∞
$(-1, 1)$	1	-1
$[-3, 4]$	4	-3
$(-10, 1]$	1	-10
$\{3, 4, 5\}$	5	3
$\{(-1)^x : x \in N\}$	1	-1

Facts About Supremum and Infimum:

Let A, B be subsets of R . Then

- (i) $\text{Sup}(A) = -\text{Inf}(-A)$
- (ii) $\text{Inf}(A) = -\text{Sup}(-A)$
- (iii) $\text{Inf}(aA) = a \text{Sup}(A), a < 0$
- (iv) $\text{Inf}(aA) = a \text{Inf}(A), a > 0$
- (v) $\text{Sup}(aA) = a \text{Inf}(A), a < 0$
- (vi) $\text{Inf}(aA) = a \text{sup}(A), a < 0$
- (vii) $\text{Sup}(aA) = a \text{Sup}(A), a > 0$
- (viii) $\text{Sup}(a + A) = a + \text{Sup}(A)$
- (ix) $\text{Inf}(a + A) = a + \text{Inf}(A)$
- (x) $\text{Sup}(A) \leq \text{Inf}(B)$ if $a \leq b \forall a \in A$ and $b \in B$

$$(xi) \quad \text{Sup } (A + B) = \text{Sup } (A) + \text{Sup } (B)$$

$$(xii) \quad \text{Inf } (A + B) = \text{Inf } (A) + \text{Inf } (B)$$

Real Sequence:

A sequence whose range set is a real number set is called a real sequence.

Bounded Sequence:

A sequence is said to be bounded if its range set is bounded.

Convergence of a Sequence:

A sequence $\{S_n\}$ is said to be converge to a number l (l as its limit) if for $\epsilon > 0$ there exists a positive integer m (depending on ϵ) such that $|S_n - l| < \epsilon$ for all $n \geq m$.

A sequence which is not convergent is called divergent sequence.

Note:

- (i) Every convergent sequence is bounded
- (ii) A sequence cannot converge to more than one limit.

Limit Points of a Sequence:

A number x_0 is said to be a limit point of a sequence $\{S_n\}$ if every neighbourhood of x_0 contains an infinite number of members of the sequence.

Thus x_0 is a limit point of a sequence if given any positive number r , $S_n \in (x_0 - r, x_0 + r)$ for an infinite number of values of n .

Note:

A limit point of the range set of a sequence is also a limit point of the sequence. But the converse may not always be true.

Following table shows the information of limit points.

Sequence	Limit Point	Range Set	Limit Point
$S_n = 1, \forall n \in \mathbb{N}$	1	$\{1\}$	No limit point
$S_n = \frac{1}{n}, \forall n \in \mathbb{N}$	0	$\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$	0
$S_n = 1 + (-1)^n, \forall n \in \mathbb{N}$	0, 2	$\{0, 2\}$	No limit point
$S_n = (-1)^n, \forall n \in \mathbb{N}$	1, -1	$\{1, -1\}$	No limit point
$S_n = (-1)^n \left(1 + \frac{1}{n}\right), \forall n \in \mathbb{N}$	1, -1	$\{ \dots, \frac{-4}{3}, -1, 1, \frac{3}{2}, \dots \}$	1, -1

Following table shows the information of some sequence and their behaviours.

Sequence	Behaviour
$\{1 + (-1)^n\}$	Oscillate
$\{n^2\}$	Divergent
$\{-2^n\}$	Divergent
$\left\{\frac{(-1)^{n-1}}{n!}\right\}$	Converges to the limit 1
$\left\{\frac{1}{n}\right\}$	Converges to limit 0
$\left\{1 + \frac{1}{n}\right\}$	Converges to limit 1

Series:

A series is a sum of terms of a sequence. If number of terms in the sum are finite, then series is called finite series otherwise infinite series.

Convergence Tests for Infinite Series:

1. A necessary condition for convergence of an infinite series $\sum a_n$ is that $\lim_{n \rightarrow \infty} a_n = 0$.
2. Let $\{S_n\}$ be a sequence of partial sums of the series $\sum a_n$. If sequence $\{S_n\}$ converges to the limit l then $\sum a_n$ also converges and $\sum a_n = l$.
3. (Cauchy Criterion) If a series $\sum a_n$ is convergent, then for every $\epsilon > 0$ there exists a positive integer N such that $|S_m - S_n| < \epsilon$ for all $m \geq N, n \geq N$. where $\{S_n\}$ sequence of partial sum.
4. If $\sum a_n$ and $\sum b_n$ are convergent series then $\sum (a_n \pm b_n)$ are also convergent.
5. (Comparison Test) Let $\sum a_n$ and $\sum b_n$ be series of positive terms with $a_n \leq b_n$ for each $n \in \mathbb{N}$. Then if $\sum b_n$ converges. So does $\sum a_n$ and if $\sum a_n$ diverges so does $\sum b_n$.
6. (Limit Comparison Test)
Let $\sum a_n$ and $\sum b_n$ be series of positive terms. Then
 - (i) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} \neq 0$, then either both the series converge or both diverge.
 - (ii) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges then $\sum a_n$ also converges.
 - (iii) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ also diverges.

7. (Cauchy Integral Test) Let $\sum a_n$ be a positive term series. If $f(n) = a_n$ is a continuous and non-increasing function on $[1, \infty[$ then $\sum a_n$ converges if $\int_1^{\infty} f(x) dx$ converges.
8. (D' Alembert's Ratio Test) Let $\sum a_n$ be a series of positive terms
- (i) If $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} < 1$, the series $\sum a_n$ converges
- (ii) if $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} > 1$, the series $\sum a_n$ diverges
9. (Cauchy's Root Test) Let $\sum a_n$ be a series of positive terms.
- (i) If $\lim_{n \rightarrow \infty} (a_n)^{1/n} < 1$, series $\sum a_n$ converges
- (ii) If $\lim_{n \rightarrow \infty} (a_n)^{1/n} > 1$, series $\sum a_n$ diverges.
10. (Leibnitz Test) Let $\sum a_n$ be an alternating series. Then $\sum a_n$ converges if $\{a_n\}$ is a non-increasing sequence and $\lim_{n \rightarrow \infty} a_n = 0$.

Absolute Convergent:

A series $\sum a_n$ is said to be absolutely convergent if the series $\sum |a_n|$ obtained on taking every term of the given series with a positive sign is convergent.

Conditional Convergent:

A series which is convergent but is not absolutely convergent is called a conditionally convergent series.

Following table shows the information of some series and their behaviours.

Infinite Series	Behaviour
$\sum \frac{1}{n}$	Divergent
$\sum \frac{1}{n^p}, p > 1$	Convergent
$\sum \frac{1}{1+n^2}$	Divergent
$\sum \frac{\tan^{-1} n}{1+n^2}$	Convergent
$\sum \frac{\sin(n\theta)}{n^2}$	Convergent

$\sum \frac{\cos(n\pi)}{n+3}$	Conditionally convergent
$\sum_{n=1}^{\infty} (-1)^n \frac{1}{n(\ln n)}$	Conditionally convergent
$\sum_{n=3}^{\infty} \frac{1}{n \log n (\log \log n)^P}, P > 1$	Convergent

Some Facts About Real Line:

Following table shows the information of some facts of real numbers.

$(\mathbb{R}, +)$	Additive Abelian group
$(\mathbb{R} - \{0\}, \times)$	Multiplicative Abelian group
$(\mathbb{R}, +, \times)$	Ring
$(\mathbb{R}, +, \times)$	Field
$\mathbb{R}(F)$	$\mathbb{R}$ is a vector space over field F where F can be $\mathbb{R}$, or $\mathbb{Q}$.
$(\mathbb{R}, d)$	$\mathbb{R}$ is a metric space with usual metric d .
$(\mathbb{R}, \tau)$	$\mathbb{R}$ is topological space with τ is a metrizable topology.
$(\mathbb{R}, \tau)$	Connected space
$(\mathbb{R}, \tau)$	Not compact
$(\mathbb{R}, \tau)$	Separable
$\mathbb{R}$	Open set
$\mathbb{R}$	Not countable
$\mathbb{R} \cup \{-\infty, \infty\}$	Extended real number system
$\mathbb{R} \cong (a, b)$ $\forall a, b, \in \mathbb{R}$ such that $a \neq b$	Homeomorphic (Topologically Equivalent)
$(\mathbb{R}, \tau)$ Where τ is a metrizable topology	T_1 – space, T_2 – space , T_3 – space , T_4 – space,
$(\mathbb{R}, \ x\)$	Normed space Where $\ x\ = x $

$(\mathbb{R}, \ x\)$	Banach space
$(\mathbb{R}, \langle x, x \rangle)$	Inner Product space where $\langle x, x \rangle = x ^2$
$(\mathbb{R}, \langle x, x \rangle)$	Hilbert space

Multiple Choice Questions:

- Let X be a set such that every non-empty subset of X has a smallest integer. Then $X =$
 (a) $\mathbb{N}$ (b) $\mathbb{Z}$ (c) $\mathbb{Q}$ (d) $\mathbb{R}$
- Which of the following statement is true:
 (a) between every two rational number, there are infinite rational number
 (b) between every two irrational number there are infinite irrational number
 (c) between every two real number there are infinite real numbers
 (d) all of these
- Which of the following is an irrational.
 (a) non-terminating reoccurring decimal (b) vulgar fraction
 (c) terminating decimal (d) square root of n where n is a prime
- If A & B are countable then which of the following is not countable set:
 (a) $A \cup B$ (b) $A \times B$ (c) $A \cap B$ (d) None
- Which of the following is countable set
 (a) $\mathbb{Z}$ (b) $\mathbb{Q}$ (c) $\mathbb{R}$ (d) None
- Let $A = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}$. Then supremum of A ($\sup A$) =
 (a) 0 (b) 1 (c) ∞ (d) 2
- Let $A = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}$. Then infimum of A ($\inf A$) =
 (a) 0 (b) 1 (c) ∞ (d) 2
- What is the supremum and infimum of empty set respectively.
 (a) $-\infty, \infty$ (b) ∞, ∞
 (c) $\infty, -\infty$ (d) $0, \infty$
- What is supremum and infimum of $\mathbb{R}$ respectively?
 (a) $\infty, -\infty$ (b) ∞, ∞
 (c) $-\infty, \infty$ (d) $0, \infty$
- What is the supremum of set $\{1, 2, 3, 4, 5, \dots, 10\}$.
 (a) 10 (b) 11 (c) 12 (d) ∞
- Which of the following set has not supremum in $\mathbb{Q}$.
 (a) $\{x \in \mathbb{Q} \mid x^2 < 2\}$ (b) $\{x \in \mathbb{Q} \mid x^2 - 2 < 4\}$
 (c) $\{x \in \mathbb{Q} \mid x^2 > -1\}$ (d) All of these
- For any two real numbers x and y which of the following is correct.
 (a) $|x + y| \leq |x| + |y|$ (b) $||x| - |y|| \leq |x - y|$
 (c) $|xy| = |x| |y|$ (d) All of these
- Let R^n be a normed space, then for $x, y \in R^n$ which of the following is true.
 (a) $\|x\| \geq 0$ (b) $\|x + y\| \leq \|x\| + \|y\|$
 (c) $\|kx\| = |k| \|x\|$ for some scalar $k \in R$ (d) All of these
- Which of the following is a Cauchy-Schwarz's inequality for $x, y \in R^n$.
 (a) $|x + y| \leq \|x\| \|y\|$ (b) $|x - y| \leq \|x\| + \|y\|$

- (c) $|xy| \leq ||x|| ||y||$ (d) $|x + y| \leq ||x|| + ||y||$
15. Which of the following is correct for $x, y, z \in \mathbb{R}^n$
- (a) $||x - y|| \geq ||x|| - ||y||$ (b) $||x - y|| < ||x - z|| + ||z - y||$
 (c) $|xy| < ||x|| ||y||$ (d) All of these
16. Let $x, y \in \mathbb{R}^n$ and $(x - y) \cdot (x + y) = 0$. Then which of the following is true.
- (a) $||x - y|| = ||x + y||$ (b) $||xy|| = ||yx||$
 (c) $||x|| = ||y||$ (d) None
17. Let A be a non-empty subset of real line then which of the following is true.
- (a) $\sup(-A) = -\inf(A)$ (b) $\sup(A) = -\inf(-A)$
 (c) $\inf(A) + \sup(-A) = 0$ (d) All of these
18. If A is a bounded set and $b > 0$, then which of the following is true.
- (a) $\sup(bA) = b \sup(A)$ (b) $\sup(b + A) = b + \sup(A)$
 (c) $\inf(b + A) = b + \inf(A)$ (d) All of these
19. The real line $\mathbb{R}$ is a metric space with metric $d(x, y) =$
- (a) $|x - y|$ (b) $\max(1, |x - y|)$
 (c) $\frac{|x - y|}{1 + |x - y|}$ (d) All of these
20. Let $x = (x_1, x_2, \dots, x_n), y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$ then
- $$\left(\sum_{i=1}^n (x_i + y_i)^2 \right)^{1/2} \leq \left(\sum_{i=1}^n x_i^2 \right)^{1/2} + \left(\sum_{i=1}^n y_i^2 \right)^{1/2} \text{ is}$$
- (a) Cauchy's inequality (b) Minkowski's inequality
 (c) Lagrang's inequality (d) All of these
21. Every normed linear space is also a
- (a) Vector space (b) Metric space
 (c) Abelian group (d) All of these
22. Let X be a normed linear space and $x, y \in X$. Then $||x - y|| =$
- (a) $||xy||$ (b) $d(x, y)$
 (c) $||x|| ||y||$ (d) $d(x - y)$
23. Which of the following is a norm on $\mathbb{R}$.
- (a) $|x|$ (b) $|x - y|$ (c) $x^2 + y^2$ (d) $|x + y|$
24. The real line $\mathbb{R}$ is
- (a) Closed (b) Open (c) Connected (d) All of these
25. Which of the following subset of $\mathbb{R}$ is connected
- (a) $\mathbb{N}$ (b) $\mathbb{Q}$ (c) $\mathbb{Z}$ (d) $[0, 1]$
26. Let X be a subset of $\mathbb{R}$ and X is connected. Then X is
- (a) Open set (b) Closed set
 (c) An interval (d) Bounded set
27. Which of the following sequence is decreasing in $\mathbb{R}$
- (a) $\{(-1)^{n+1}\}$ (b) $\left\{ \left(1 + \frac{1}{n} \right) \right\}$

- (c) $\left\{1 - \frac{1}{n}\right\}$ (d) $\{(2n + 1)\}$
28. A sequence $\{x_n\}$ in R is increasing if
- (a) $x_n - x_{n+1} \geq 0$ (b) $\frac{x_n}{x_{n+1}} \geq 1$
- (c) $\frac{x_{n+1}}{x_n} \leq 1$ (d) $\frac{x_{n+1}}{x_n} \geq 1$
29. Which of the following sequence is convergent.
- (a) $\frac{1}{n}$ (b) $\frac{1}{\sqrt{n}}$ (c) $\frac{1}{n^2}$ (d) All of these
30. Which of the following sequence is convergent to 3.
- (a) $\left\{\frac{n+3}{n}\right\}$ (b) $\left\{\frac{n+1}{n^2+3}\right\}$
- (c) $\left\{\frac{n(3n+2)}{n^2-7}\right\}$ (d) $\left\{\frac{n^2+3}{3}\right\}$
31. Which of the following sequence is convergent to 1.
- (a) $\left\{\frac{1}{n}\right\}$ (b) $\left\{2 - \frac{1}{n}\right\}$ (c) $\{(n)^{1/n}\}$ (d) $\left\{\frac{n+2}{n^2-1}\right\}$
32. "Every bounded sequence has a convergent subsequence" is the statement of:
- (a) Cauchy Theorem (b) Bolzano-Weierstrass Theorem
- (c) Nested Interval Theorem (d) Taylor's Theorem
33. Let $\{x_n\}$ be a sequence such that $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf x_n = \lim_{n \rightarrow \infty} \sup x_n$. Then $\{x_n\}$ is
- (a) Bounded sequence (b) Cauchy sequence
- (c) Convergent sequence (d) Increasing sequence
34. Which of the following is true for real line R .
- (a) R is complete (b) R is connected
- (c) R is not compact (d) All of these
35. Closed and bounded subsets of R^n are
- (a) Connected (b) Compact
- (c) Finite (d) Complete
36. Let A be a compact subset of a metric space. Then
- (a) A is totally bounded (b) A is complete
- (c) Every infinite subset of A has limit point in A
- (d) All of these
37. Which of the following is a topological property.
- (a) Compactness (b) Connectedness
- (c) Completeness (d) All of these
38. A series $\sum x_n$ is divergent if
- (a) $|x| < 1$ (b) $\lim_{n \rightarrow \infty} x_n = 0$

- (c) $|x| = 0$ (d) $|x| \geq 1$
39. A series $\sum x_n$ is convergent if $\lim_{n \rightarrow \infty} x_n =$
(a) -1 (b) 0 (c) 1 (d) ∞
40. The series $\sum \frac{1}{n^p}$ is convergent if
(a) $p = 0$ (b) $p > 1$ (c) $p < 1$ (d) $p \geq 2$
41. Which of the following infinite series is divergent
(a) $\sum \frac{1}{(2n-1)(2n+1)}$ (b) $\sum \left(\frac{1}{2^n} - \frac{1}{3^n} \right)$
(c) $\sum \left(1 + \frac{1}{n^2} \right)$ (d) $\sum \frac{2^n - 1}{3^n}$
42. Which of the following series is divergent
(a) $\sum \frac{n^2 + 1}{n^2}$ (b) $1 + \frac{(-1)^n}{1+n}$
(c) $\frac{1 + (-1)^n}{1 - (-1)^n}$ (d) None
43. The infinite series $\sum \frac{1}{n!}$ converges to
(a) 0 (b) 1 (c) e (d) 3
44. Which of the following infinite series is divergent
(a) $\sum \frac{1}{n} \sin^2 \frac{x}{n}$ (b) $\sum \frac{n!}{n}$
(c) $\sum \frac{n^3 + 3n + 1}{5n^5 + n + 1}$ (d) $\sum \frac{1}{n^2}$
45. The positive term series $\sum x_n$ is convergent if $\lim_{n \rightarrow \infty} n \log \frac{x_n}{x_{n+1}} = l$
(a) $l = 1$ (b) $l > 1$ (c) $l < 1$ (d) $l = 0$
46. Let $a_n > 0, b_n > 0$. If $\sum a_n$ & $\sum b_n$ converges, then which of the following also converges.
(a) $\sum (a_n)^2$ (b) $\sum \sqrt{a_n b_n}$
(c) $\sum \frac{a_n}{1 + a_n}$ (d) All of these
47. Every absolutely convergent series is:
(a) Bounded (b) Cauchy
(c) Convergent (d) All of these

48. Which of the following series is divergent:

- (a) $\frac{1}{1} + \frac{1}{2} - \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \frac{1}{6} - \frac{1}{7} - \frac{1}{8} + \dots$ (b) $\frac{1}{2^2} - \frac{2^2}{3^3} + \frac{4^4}{5^5} + \dots$
 (c) $1 + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{5}} + \frac{1}{\sqrt{7}} - \frac{1}{\sqrt{4}} + \dots$ (d) $1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!}$

49. The series $\sum_{n=0}^{\infty} \frac{x^n}{n^2}$ converges absolutely if

- (a) $|x| < 1$ (b) $|x| > 1$ (c) $|x| \geq 1$ (d) $|x| \leq 1$

50. If radius of convergent of series $\sum a_n x^n$ is p . Then radius of convergent of $\sum n a_n x^{n-1}$ is

- (a) $p - 1$ (b) p (c) $p + 1$ (d) $p - 2$

51. Which of the following series is convergent.

- (a) $\sum_1^{\infty} \frac{(2n)!}{4^n}$ (b) $\sum_1^{\infty} \frac{2^n}{n(n+2)}$
 (c) $\sum_1^{\infty} \frac{1}{n^n}$ (d) $\sum_1^{\infty} \frac{3^n}{n^3}$

52. Which of the following series is divergent.

- (a) $\sum_1^{\infty} n \left(\frac{\pi}{n}\right)^n$ (b) $\sum_1^{\infty} \left(\frac{e}{\ln n}\right)^n$
 (c) $\sum_1^{\infty} \frac{1^n + 2^n}{3^n}$ (d) $\sum_1^{\infty} \left(\frac{5n}{2n+1}\right)^{3n}$

53. Which of the following alternating series is convergent.

- (a) $\sum_1^{\infty} (-1)^{n-1} \frac{n^2}{n^2 + 1}$ (b) $\sum_1^{\infty} (-1)^{n-1} \frac{1}{e^n}$
 (c) $\sum_1^{\infty} (-1)^n \frac{\ln n}{n}$ (d) $\sum_1^{\infty} (-1)^n n \tan\left(\frac{1}{n}\right)$

54. The radius of convergence of series $\sum_{n=0}^{\infty} (-1)^n 2^n \sin^n x$ is

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

55. The radius of convergence of series $\sum_{n=1}^{\infty} \frac{n^n}{n!} x^n$ is:

- (a) 1 (b) 2 (c) e (d) e^{-1}

56. The interval of convergence of series $\sum_1^{\infty} x^{2n}$ is:

- (a) $(-2, 2)$ (b) $(-1, 1)$ (c) $[-1, 1]$ (d) $(1, \infty)$

57. The interval of convergence of series $\sum_{n=2}^{\infty} \frac{x^n}{(\ln n)^n}$

- (a) $(-e, e)$ (b) $(-1, 1)$ (c) $(-\infty, \infty)$ (d) $(0, \infty)$

58. The interval of convergence of series $\sum_{n=1}^{\infty} n^2 (x-2)^n$ is:
(a) $(0, 1)$ (b) $(-1, 1)$ (c) $(1, 3)$ (d) $(0, \infty)$
59. If domain $f(x) = \text{Range } f(x) = R$, then $f(x)$ is:
(a) rational function (b) piece wise function
(c) linear function (d) all of these
60. The range of absolute valued function is:
(a) R (b) R^+ (c) $[0, \infty[$ (d) $(-\infty, \infty)$
61. If a graph of a function is u-shaped then function is:
(a) rational function (b) linear function
(c) quadratic function (d) cubic function
62. The graph of function $f(x) = ax^2 + bx + c$ is u-shaped if
(a) $a = 0$ (b) $a > 0$ (c) $a < 0$ (d) none
63. The graph of a function $f(x) = ax^2 + bx + c$ is n-shaped if
(a) $a = 0$ (b) $a > 0$ (c) $a < 0$ (d) none
64. The inverse $f^{-1}(x)$ exists if:
(a) $f(x)$ is surjective (b) $f(x)$ is one to one
(c) $f(x)$ is bijective (d) $f(x)$ is continuous
65. If $f(x)$ is a step function then $f(5) =$
(a) 1 (b) 4 (c) 5 (d) -5
66. If $f(x)$ is a step function then $f(-3) =$
(a) -4 (b) -3 (c) -2 (d) 0
67. Let f and g be two real valued function such that their domains are A and B respectively. Then $A \cap B$ is a domain of:
(a) $f + g$ (b) $f \circ g$ (c) f / g (d) all of these
68. Let f and g be two real valued functions such that supremum of f and g exists in R . Then which of the following is true.
(a) $\text{Sup}(f + g) \leq \text{Sup}(f) + \text{Sup}(g)$ (b) $\text{Sup}(fg) \leq \text{Sup}(f) \text{Sup}(g)$
(c) $|\text{Sup}(f)| \leq \text{Sup}|f|$ (d) all of these
69. Let f and g be two real valued functions such that infimum of f and g exists in R . Then which of the following is true.
(a) $\inf(f + g) \geq \inf(f) + \inf(g)$ (b) $\inf(fg) \geq \inf(f) \cdot \inf(g)$
(c) $|\inf(f)| \geq \inf|f|$ (d) all of these
70. Let f be a real valued function such that $\text{Sup}|f| - \inf|f| \leq \text{Sup}(f) - \inf(f)$. Then f is:
(a) bounded above (b) bounded below
(c) bounded function (d) none
71. Let f and g be bounded real valued functions, then which of the following is correct.
(a) $\text{Sup}(f) + \inf(f) \leq \text{Sup}(f + g)$ (b) $\text{Sup}(f) + \inf(g) \leq \text{Sup}(f + g)$
(c) $\text{Sup}(f) + \inf(g) \geq \inf(f + g)$ (d) $\text{Sup}(f) - \inf(g) = \text{Sup}(f + g)$
72. Which of the following function is convex on $[0, 1]$.
(a) $f(x) = x + 1$ (b) $f(x) = x^3 + x^2 + x$

- (c) $f(x) = |x|$ (d) $f(x) = \log(x)$
73. Which of the following function is continuous on $[0, \infty]$:
(a) $\sin(x)$ (b) $\tan(x)$
(c) $\sec(x)$ (d) all of these
74. Let f be a continuous function on $[a, b]$ such that $f(a) < 0$ and $f(b) > 0$ then there exists $c \in [a, b]$ such that
(a) $f(c) > 0$ (b) $f(c) < 0$
(c) $f(c) = 0$ (d) none
75. Let $f \in C[a, b]$ and $f(a) \neq f(b)$ then there is λ such that $f(a) < \lambda < f(b)$ then for $a < c < b$,
(a) $f(\lambda) = c$ (b) $f(c) = \lambda$
(c) $f(c) > \lambda$ (d) $f(c) < \lambda$
76. If $f \in C[a, b]$ then:
(a) f is differentiable (b) f is bounded
(c) f is linear (d) f is integrable
77. Let f be differentiable on $[a, b]$. Then $f'(x) = 0 \forall x \in [a, b]$ if:
(a) $f(x)$ is continuous (b) $f(x)$ is constant
(c) $f(x)$ is increasing (d) none
78. Let $f \in C[a, b]$ and $f'(x) \geq 0$ then:
(a) $f(x)$ is increasing (b) $f(x)$ is decreasing
(c) $f(x)$ is monotonically increasing (d) $f(x)$ is monotonically decreasing
79. Let $f \in C[a, b]$ and $f'(x) \leq 0$, then
(a) $f(x)$ is increasing (b) $f(x)$ is decreasing
(c) $f(x)$ is monotonically increasing (d) $f(x)$ is monotonically decreasing
80. If f is continuous on $[a, b]$ and differentiable on $]a, b[$, then there exists $c \in]a, b[$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$ is a statement of
(a) Rolle's theorem (b) Cauchy's theorem
(c) Lagrange's theorem (d) Riemann's theorem
81. $[\text{gof}(x)]' =$
(a) $g'(f(x))$ (b) $g'(f'(x)) f(x)$
(c) $g(f(x)) f'(x)$ (d) $g(f(x)) f'(x)$
82. Let $f(x)$ be differentiable function, then $f(x)$ is strictly increasing on an open set X if for some $c \in X$,
(a) $f'(c) = 0$ (b) $f'(c) > 0$
(c) $f'(c) < 0$ (d) none
83. If $f'(c) < 0$ for some $c \in X$, where X is an open interval, then $f(x)$ is:
(a) increasing (b) strictly increasing
(c) strictly decreasing (d) decreasing
84. Let $f(x)$ be continuous on $[a, b]$ and differentiable on $]a, b[$ such that $f(a) = f(b)$, then by Rolle's theorem, there is $c \in]a, b[$ such that

- (a) $f'(c) = 0$ (b) $f'(c) > 0$
 (c) $f'(c) < 0$ (d) $f(c) = f(b) - f(a)$
85. In a Taylor's series of $f(x)$, the remainder after n terms $R_n(x) = f^{(n+1)}(c) \frac{(x-x_0)^{n+1}}{(n+1)!}$ is
 (a) Lagrange's remainder (b) Taylor's remainder
 (c) Cauchy's remainder (d) none
86. In a Taylor's series of $f(x)$ the remainder after n term $R_n(x) = f^{(n+1)}(c) \frac{(x-c)^n (x-x_0)}{n!}$ is
 (a) Lagrange's remainder (b) Cauchy's remainder
 (c) Taylor's remainder (d) none
87. The infinite series of $f(x)$, $f(x) = f(x_0) + f'(x_0) \frac{(x-x_0)}{1!} + f''(x_0) \frac{(x-x_0)^2}{2!} + \dots$ for $x_0 = 0$ is a
 (a) Taylor's series (b) Maclaurin's series
 (c) Lagrange's series (d) none
88. A function $f(x)$ is Riemann integrable on $[a, b]$ if $f(x)$ is
 (a) symmetric on $[a, b]$ (b) continuous on $[a, b]$
 (c) constant on $[a, b]$ (d) uniformly continuous on $[a, b]$
89. If $f(x)$ is Riemann integrable over $[a, b]$, then:
 (a) $\left| \int_a^b f(x) dx \right| = 0$ (b) $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$
 (c) $\left| \int_a^b f(x) dx \right| = \int_a^b |f(x)| dx$ (d) $\left| \int_a^b f(x) dx \right| \geq \int_a^b |f(x)| dx$
90. If $f(x)$ is integrable on $[a, b]$, then which of the following is also integrable on $[a, b]$
 (a) $\sqrt{f(x)}$ (b) $\log f(x)$
 (c) $(f(x))^2$ (d) $\sin(f(x))$
91. Which of the following integral is convergent.
 (a) $\int_0^\infty \frac{\tan^{-1} x}{1+x^2} dx$ (b) $\int_1^\infty x^{-2} e^{-x} dx$
 (c) $\int_0^\infty \frac{\sin x}{(1+x)^2} dx$ (d) all of these
92. Which of the following is divergent.
 (a) $\int_1^\infty e^{-x^2} dx$ (b) $\int_0^1 \frac{x}{\sqrt{1-x^2}} dx$ (c) $\int_0^\infty \frac{\sin x}{x} dx$ (d) none

KEY To MCQ Of Real Analysis

1	a	2	d	3	d	4	c	5	c	6	b
7	a	8	a	9	a	10	a	11	d	12	d
13	d	14	c	15	d	16	c	17	d	18	d
19	d	20	B	21	d	22	b	23	a	24	d
25	d	26	C	27	b	28	d	29	d	30	c
31	c	32	B	33	c	34	d	35	b	36	d
37	d	38	D	39	b	40	b	41	c	42	d
43	c	44	B	45	b	46	d	47	c	48	c
49	a	50	B	51	c	52	d	53	c	54	d
55	d	56	B	57	c	58	c	59	c	60	c
61	c	62	B	63	c	64	b	65	c	66	b
67	d	68	D	69	d	70	c	71	a	72	c
73	a	74	C	75	b	76	b	77	b	78	c
79	d	80	B	81	c	82	b	83	c	84	a
85	a	86	B	87	b	88	b	89	b	90	c
91	d	92	C								